SELF-SUPERVISED LEARNING

Learning without labels

- Every label needs an expert. A segmentation mask can take a doctor an hour for one scan.
- So labelled medical datasets stay small: a thousand images is a big study. ImageNet has 1.2 million.
- But hospitals already store hundreds of thousands of scans that nobody has labelled.
- The idea: make up a task the scan can **check by itself**. Then the whole archive becomes training data, with no one labelling anything.
- It is catching on fast: medical imaging papers using this idea grew more than tenfold from 2019 to 2021.

Huang et al., *Self-supervised learning for medical image classification: a systematic review and implementation guidelines*, npj Digital Medicine

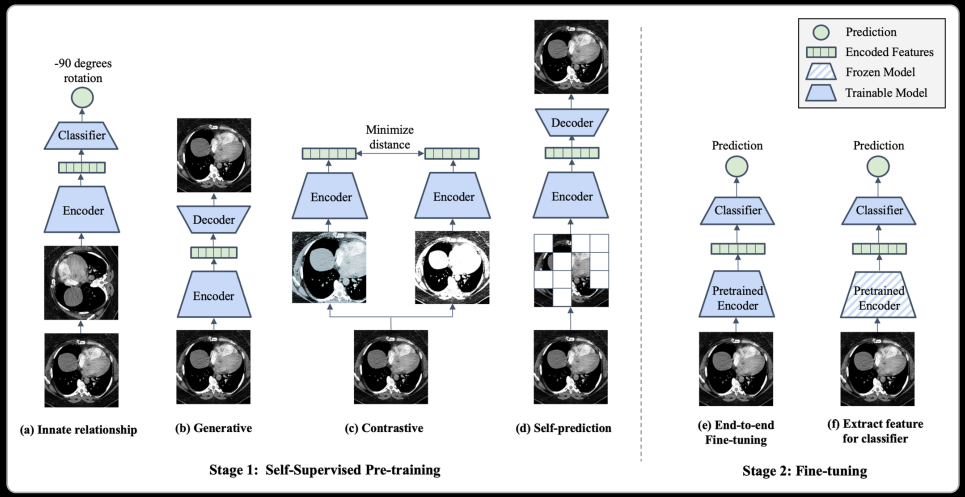
What is Self-Supervised learning?

- Supervised learning: train the model directly on the task you care about, using ground truth labels: class labels, expert masks, outcomes from the record.
- Self-supervised learning: do *not* train on the task directly.
  - Invent a **pretext task** whose answer is generated from the raw data itself.
  - Use it to learn a good feature encoder. No external labels are used at this stage.
  - Then fine-tune that encoder on the real task, with far fewer labels.

What Are the Advantages of Self-Supervised Learning?

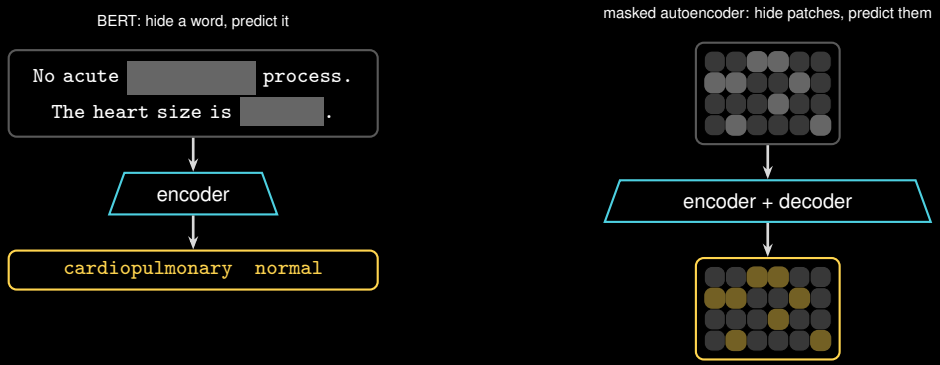
- Scarce labels: where labelling needs an expert, the label is the bottleneck. Self-supervision puts to work the data that would otherwise sit unused for want of a label.
- Data at scale: training scales with the archive, not with how much annotation you can buy.
- Versatile: the same recipe learns from text, speech and images. What comes out is a general encoder, not a solution to one task.
- Robust and general: trained on broader data, the model overfits less. That matters when deployment is diverse and unpredictable.

FOUR FAMILIES OF PRETEXT TASK



Huang et al., npj Digital Medicine (CC BY 4.0)

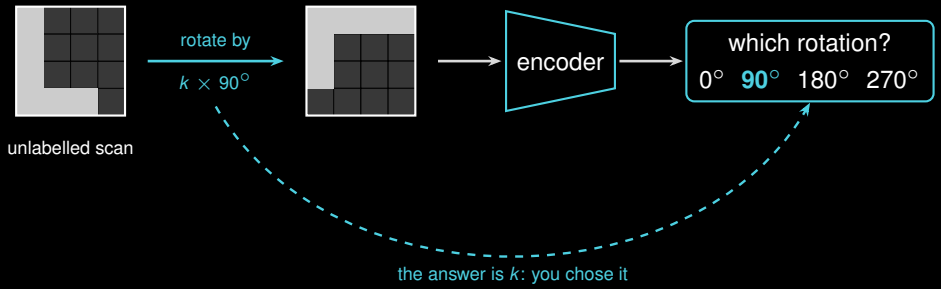
YOU ALREADY USE ONE EVERY DAY



BERT, the network behind web search since 2019, was trained by hiding words and predicting them; GPT predicts the next one. **The sentence is its own label.** The same trick on pixels is a masked autoencoder, and the same trick on a chest x-ray is a pretext task nobody had to annotate.

"Fill in the blanks" framing: Y. LeCun, via IBM

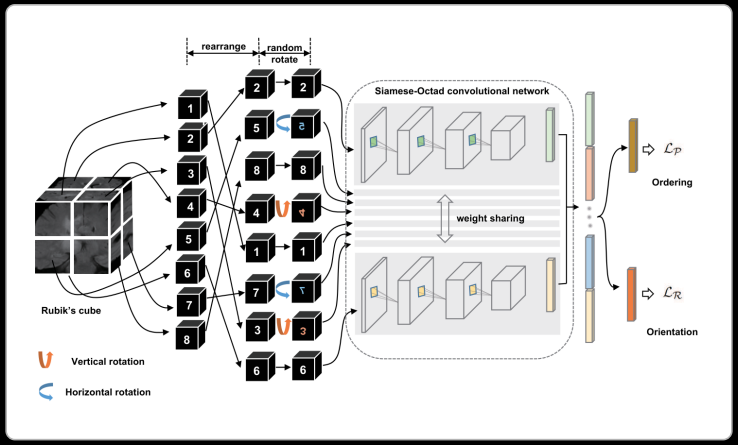
INNATE RELATIONSHIP: PREDICT THE TRANSFORMATION



- Make a change you chose: rotate the scan, shuffle it into a jigsaw, or reorder its slices.
- Train a classifier to name the change. **The label is free**, because you made the change.
- To answer, the network has to know what upright, correctly ordered anatomy looks like. That knowledge is the representation.

Taxonomy after Huang et al., npj Digital Medicine

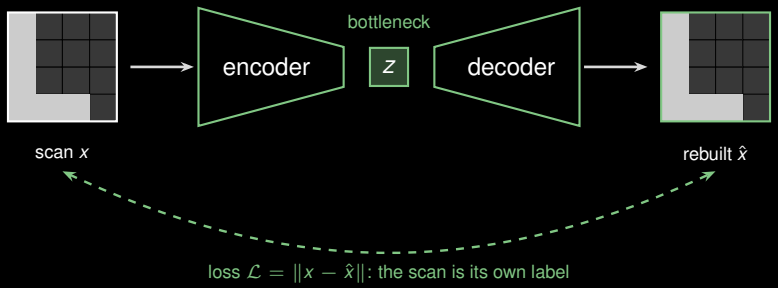
INNATE RELATIONSHIP: RUBIK'S CUBE RECOVERY



Cut a 3D scan into a $2 \times 2 \times 2$ grid of sub-cubes, **shuffle** them and **rotate** some, then train a network to recover the original **order** and **orientation**: two losses, no labels. Fine-tuning from it beats training from scratch on brain hemorrhage classification and brain tumour segmentation.

Zhuang et al., *Self-supervised Feature Learning for 3D Medical Images by Playing a Rubik's Cube*, MICCAI

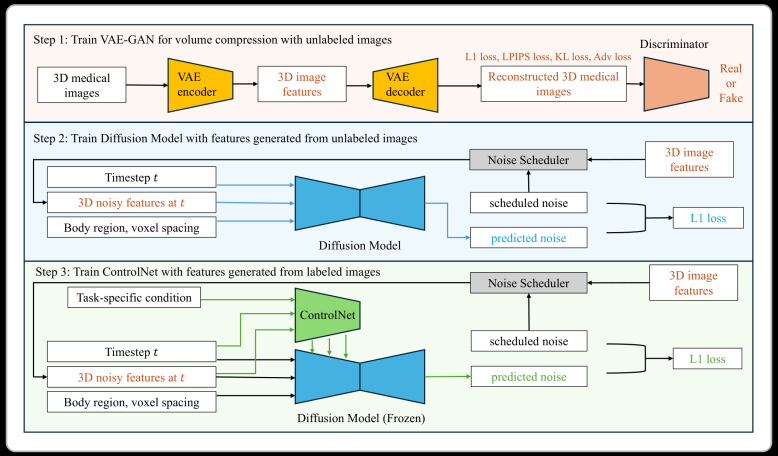
GENERATIVE: REBUILD THE WHOLE INPUT



- Squeeze the **whole** scan through a narrow bottleneck, then rebuild all of it. The loss compares the rebuild with the input.
- The bottleneck z is too small to copy the scan, so it has to **summarise** it. That summary is the representation.
- Rare as a pretext task for classification (3 of 79 studies), but it is the first stage of today's medical image generators.

Taxonomy after Huang et al., npj Digital Medicine

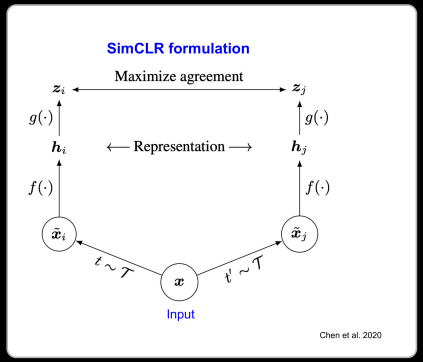
GENERATIVE: WHAT MAISI LEARNS FIRST



Step 1: an autoencoder learns to compress and rebuild 3D scans, from 37,243 CT and 17,887 MRI volumes with **no labels**. **Step 2**: a diffusion model learns to generate in that compressed space, still unlabelled. Labels enter only at **step 3**, to steer it. In Module 2 we used MAISI to make data; this is what it had to learn first.

Guo et al., *MAISI: Medical AI for Synthetic Imaging*, WACV (CC BY 4.0)

SimCLR: the shape of a contrastive method



Four moving parts

1. One image x, two random transformations t, t', giving two views $\tilde{x}_i$ and $\tilde{x}_j$.
2. A **shared** encoder f produces representations h_i, h_j.
3. A small projection head g maps those to z_i, z_j.
4. A loss that pulls z_i and z_j together and pushes every other view in the batch away.

The encoder never learns what the image *is*. It only learns what survives a transformation.

Chen et al., *A Simple Framework for Contrastive Learning of Visual Representations*, ICML

The contrastive loss is cross-entropy in disguise

$$\ell_{i,j} = -\log \frac{\exp\big(\mathrm{sim}(z_i, z_j)/\tau\big)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp\big(\mathrm{sim}(z_i, z_k)/\tau\big)} \qquad \mathrm{sim}(u, v) = \frac{u^\top v}{\|u\| \, \|v\|}$$

Read it as a classifier

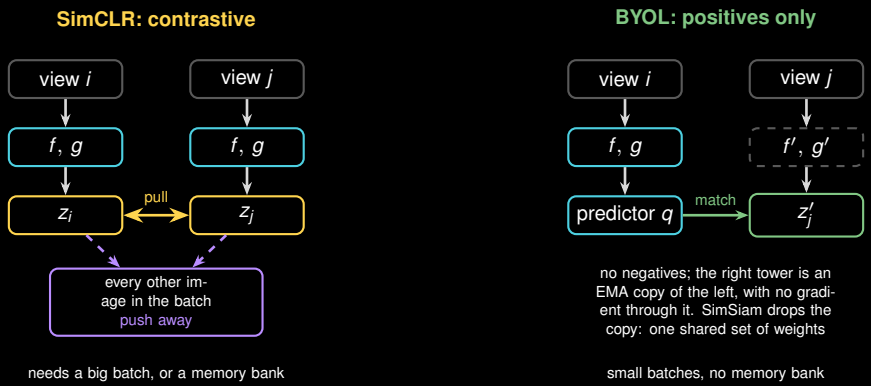
Given view i, its partner j must score higher than all $2N - 2$ other views in the batch. That is a softmax over $2N - 1$ candidates, and the "correct class" is the other view of the same image.

The two knobs

Cosine similarity compares direction, not magnitude. The **temperature** τ sets how peaked the distribution is, and it matters more than it looks.

Same negative log-likelihood we minimized in Module 1. Only the notion of "class" has changed.

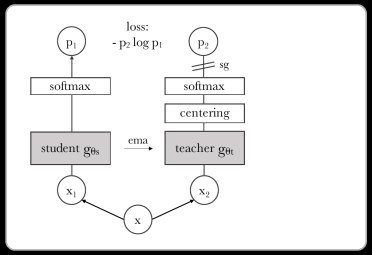
Do you even need the negatives?



No negatives, and no loss: BYOL reaches 74.3% on an ImageNet linear probe to SimCLR's 70%. The invariant is **"two views of one scan should agree"**; negatives were only ever there to stop the encoder answering every input with the same vector. BYOL blocks that with predictor, stop-gradient and EMA target together; SimSiam showed the EMA can go if the other two stay. The method is a detail. The invariant is the idea.

Grill et al., *Bootstrap Your Own Latent*, NeurIPS; Chen & He, *SimSiam*, CVPR

DINO: DISTILLATION WITH NO TEACHER TO DISTIL FROM



Two views of x; the student is trained, the teacher is an exponential moving average of it.

BYOL, with one change

The output is no longer a vector but a **softmax over K prototypes**, and the loss is **cross-entropy** between the teacher's distribution and the student's: $-p_t \log p_s$. Same shape as knowledge distillation, except the teacher is built from the student.

Two knobs against collapse

Centering: subtract the batch mean from the teacher's logits, so no single prototype can win everything. **Sharpening**: a low teacher temperature, so it cannot answer "all prototypes equally" either. One pushes towards uniform, the other away; together they hold.

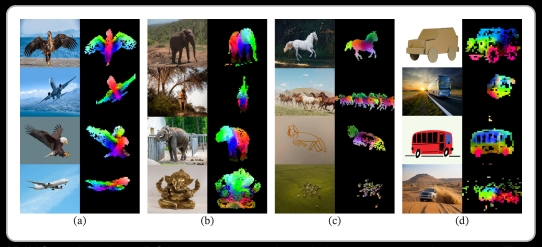
Third time the same loss. SimCLR, the classifier from Module 1, and now DINO all minimize a cross-entropy; only the "correct class" keeps changing. Trained this way with **no labels**, DINO ViT-S/8 reaches **78.3%** on ImageNet with a nearest-neighbour classifier; the larger ViT-B/8 reaches **80.1%** with a linear probe.

Caron et al., *Emerging Properties in Self-Supervised Vision Transformers*, ICCV; adapted from CS231n

What emerges when nobody says what to look at



Self-attention of the class token, last layer, DINO ViT-S/8. No mask, no box, no label was ever shown to this network.



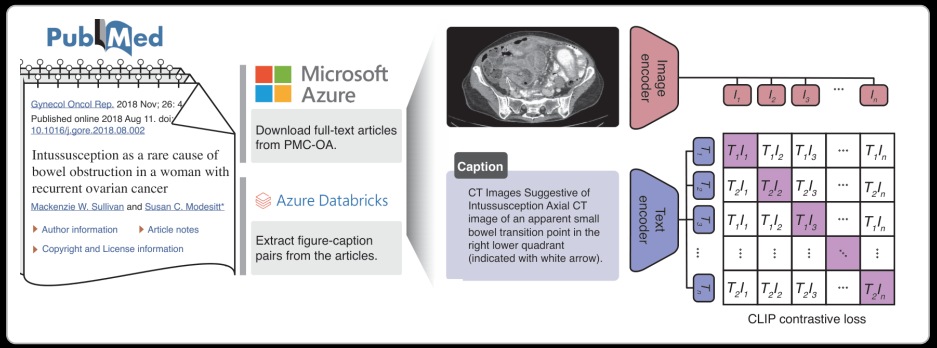
DINOv2: first three PCA components of patch features, one colour each. The same part gets the same colour across images.

Why this matters here

The encoder learned **where the object is** and **which part is which** from the pretext task alone. DINOv2 scaled the recipe to a 1.1B ViT on 142M curated images, then distilled it down; its frozen features are the off-the-shelf encoder much of today's work starts from, medical imaging included.

Caron et al., ICCV; Oquab et al., *DINOv2: Learning Robust Visual Features without Supervision*, TMLR

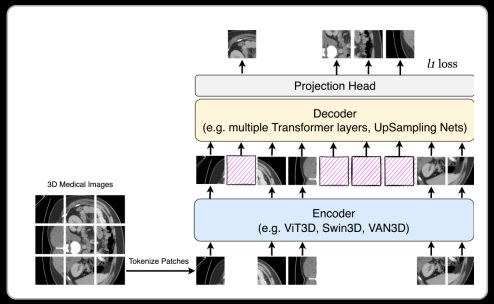
CONTRASTIVE: BIOMEDCLIP, AN IMAGE AND ITS CAPTION



The same contrastive loss, with one change: the positive pair is **an image and its caption**, not two views of one image. 15 million figure and caption pairs from 4.4 million PubMed Central articles; each image is pulled towards its own caption and pushed away from every other caption in the batch. It beats the radiology-specific BioViL on RSNA pneumonia detection. Pairing images with text is the whole of **next lecture**.

Zhang et al., *BiomedCLIP: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs*, arXiv

OR JUST HIDE PATCHES: MASKED IMAGE MODELING



MAE on a 3D CT volume: only the visible patches enter the encoder; a light decoder fills in the hatched ones.

The recipe

Cut the scan into patches. Hide **75%** of them at random. Encode only the rest, reconstruct the hidden voxels, and take the loss **on the hidden patches only**. Throw the decoder away.

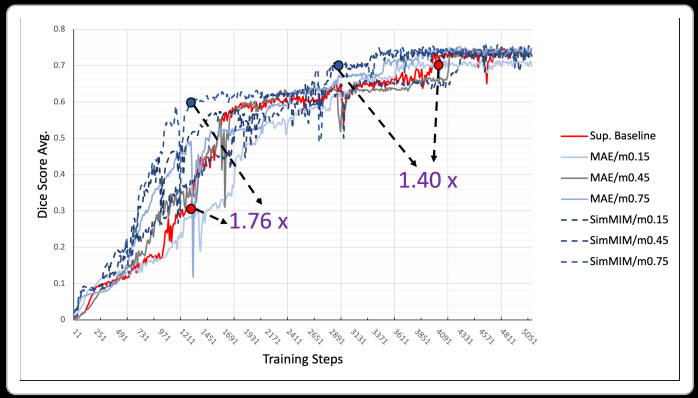
Why hide so much

Images are redundant. Hide 20% and the network copies the neighbours; hide 75% and it has to know where the kidney is to draw it.

Note the reversal of the contrastive slides' worry: here the hidden lesion is the **answer**, not a nuisance to be invariant to.

Chen et al., *Masked Image Modeling Advances 3D Medical Image Analysis*, WACV; He et al., *MAE*, CVPR; Xie et al., *SimMIM*, CVPR

SELF-PREDICTION: MASKED PRE-TRAINING ON CT

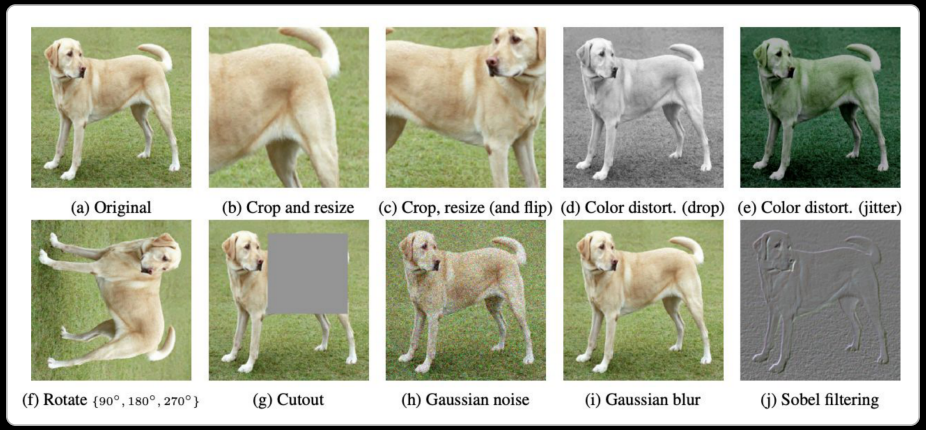


Pre-train with MAE or SimMIM on CT, then fine-tune for multi-organ segmentation. Dice reaches **0.75 to 0.76**, where contrastive SimCLR reaches 0.72. And the curves: masked pre-training (blue) reaches Dice 0.70 **1.4× sooner** than supervised training from scratch (red).

Chen et al., *Masked Image Modeling Advances 3D Medical Image Analysis*, WACV

The medical catch

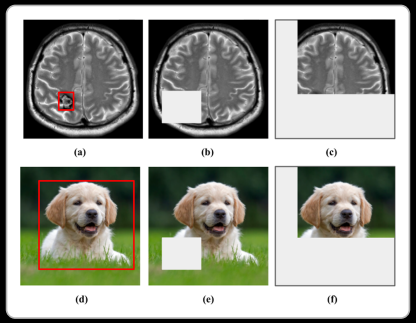
Ask the room: which of these are safe on a chest x-ray?



SimCLR's whole result rests on the transformation set: the encoder learns exactly what those transformations tell it to ignore. Crop and resize, colour distortion, rotation, cutout, Gaussian noise, blur, Sobel filtering. **Which of them would you allow on a radiograph?**

Chen et al., ICML

WHICH OF THESE ARE SAFE ON A CHEST X-RAY?



T2 MRI with a cavernoma (red), then masked, then cropped.

The dog survives both. The lesion survives neither.

Usually fine

Small crops, mild blur or noise, small rotations, **mild** brightness and contrast: real acquisition variability. Hard contrast can wash out a faint finding.

Meaningless

Colour drop and hue jitter on **one channel**. The transformation the paper found most important is not available here.

Dangerous

Horizontal flip teaches that **left and right do not matter**. Large crops and random masks can **delete the finding**, as on the left.

Scans of one protocol all look **alike**, so the class is not “the main object”, and the finding is **small and local**, so a crop that would never lose a dog can lose a tumour. In supervised training a bad augmentation costs accuracy; in a pretext task it defines what the representation may encode.

Huang et al., npj Digital Medicine (CC BY 4.0)

BUILD POSITIVES FROM THE CLINIC

adjacent slices
of one CT series

two views
of one study: PA and lateral, CC and MLO

one patient
scanned again a year later

image + its report
next lecture

each is a “same thing, different view” pair that already exists in PACS, with no transformation invented by you

Why it is safer

A neighbouring slice or a later scan still contains the lesion. The invariance you teach is “same anatomy, same patient”, which is the one you actually want.

Why it works better

Across the review, pretext tasks built on clinical structure improved downstream results by **13.7%** over supervised baselines; generic transformation tasks by **5.0%**.

Huang et al., npj Digital Medicine